Sanaul Haque

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RESEARCH PROPOSAL

Coupled Electrodynamic and Thermal Limits of Learning in Dense 3D Memristive Crossbar Networks

Parasitic capacitance, inductive crosstalk, state drift, and thermal runaway

Full Legal Name	Yoochan Chae
Institution Name	Gems American Academy Abu Dhabi
Category	Junior
Class/Grade/Year	12 (Grade)
Country	United Arab Emirates
Major/Subjects	Computational Physics & Engineering
Registered Email Address	woojja0225@gmail.com
Submission Date	15/09/26

Research Topic

Investigation of coupled electrodynamic interactions and thermodynamic heat dissipation in dense 3D memristive crossbar networks

Abstract

Dense three-dimensional memristive crossbars can store synaptic weights and perform vector-matrix multiplication in the same structure, but the operation of one cell changes the electrical and thermal conditions of nearby cells. This proposal will build a coupled model that treats each interconnect segment as a distributed resistance-inductance-capacitance network and each memristor as a temperature-dependent state element. A thermal network will return local Joule heating to the electrical model at every time step. The study will then test how cell pitch, layer count, pulse rise time, duty cycle, state distribution, and heat-sink resistance affect unintended conductance drift, the loss of a stable thermal operating point, and neural-network training. Published device and array measurements will be used for calibration rather than new fabrication. The main output will be a stability map that separates reliable operation, recoverable drift, and runaway-prone regimes, together with pulse and mapping rules that preserve learning accuracy. The work is intended to explain when electrical parasitics and heat can be treated separately and when that simplification fails.

1. Research Problem and Rationale

A memristive crossbar represents a matrix by the conductance at each row-column junction. Applied row voltages produce column currents, so Ohm's and Kirchhoff's laws perform the multiply-accumulate operation in parallel [1]. Three-dimensional stacking increases the number of weights above a given chip area; working devices with multiple crossbar layers have already been demonstrated [2]. The same geometry also places long, closely spaced conductors and heat-generating filaments inside a small volume. Voltage pulses therefore do not arrive at every cell with their commanded shape. Line resistance creates position-dependent loss, while capacitance and self- or mutual inductance change edge timing, overshoot, and crosstalk [3]-[6].

The electrical error feeds a second mechanism. Switching and leakage generate Joule heat, and the switching rate of filamentary RRAM is strongly temperature dependent [7]. Heat spreads through electrodes and dielectric layers into nearby cells. Numerical work on 3D RRAM found layer-dependent temperatures and retention loss in disturbed cells [8], while later finite-element work showed that pulse spacing can preserve heat from one update into the next [9]. Recent measurements also found that higher temperature compresses the usable conductance range of a 1T1R synapse and lowers online-learning accuracy [10]. In a stacked network, these effects share variables: a crosstalk pulse changes current, current changes temperature, and temperature changes both resistance and the next state update. A model that adds independent "electrical noise" and "thermal noise" after the circuit calculation cannot represent this feedback loop.

This matters most during in-situ learning. Training repeatedly programs selected cells while neighboring cells are half-selected or idle. If the error is spatially correlated, the update rule no longer sees random device noise; it sees a bias tied to layer, distance from the drivers, and recent activity. The proposed research will locate the operating conditions at which that bias becomes large enough to cause persistent state drift or an unstable electrothermal response.

2. Existing Literature and Research Gap

Device studies provide the components of a coupled model. Temperature-activated filament growth explains how set and reset transitions depend on voltage and time [7]. Parasitic capacitance can prolong current overshoot without substantially changing its peak [6], and models with added capacitance and inductance reproduce non-ideal terminal behavior [11]. The state equation must therefore use the voltage and temperature experienced by the device, not the command at the driver.

Array studies now include more than line resistance. PEEC models extract resistance, capacitance, self-inductance, and mutual inductance, then reproduce crosstalk and delay without a full electromagnetic solve at every step [4]. S-parameter work also represents delayed mutual coupling [5]. Wang et al. combined interconnect parasitics with an electrothermal model for a 1024 by 1024 array [3]. A related study measured recognition loss in a two-layer 197 by 40 by 20 network [12]. Neither resolves vertical heat paths and interlayer coupling during learning in a dense 3D stack.

Thermal studies address the vertical geometry. Sun et al. found heat transfer into disturbed cells in a 3 by 3 by 3 block [8]. Schön and Menzel coupled a valence-change model to finite-element heat flow and identified spatial and temporal correlations under pulse trains [9]. As layer count rises, selector position and the path to a cooling surface also affect peak temperature [13]. The missing experiment varies RLC coupling, thermal boundaries, 3D layer position, and learning activity together, then locates a stability boundary rather than one peak-error case.

Learning demonstrations provide the endpoint. Metal-oxide crossbars have supported in-situ perceptron training [14], larger arrays have implemented convolutional inference and hybrid training [15], and self-rectifying arrays have linked cell coupling to recognition accuracy [16]. The proposed study will test whether the resulting error is random, spatially correlated, or self-reinforcing.

3. Research Question and Hypotheses

The central research question is: Under what combinations of 3D array density, interconnect coupling, pulse timing, activity pattern, and thermal boundary resistance do parasitic capacitance, inductive crosstalk, and Joule heating change from tolerable perturbations into correlated state drift, electrothermal runaway, and measurable degradation of in-situ learning?

H1. At slow pulse edges, line resistance and capacitance will dominate waveform error; as rise time approaches the interconnect delay, mutual inductance will add layer-dependent overshoot and nonselected-cell disturbance.

H2. The transition to persistent drift will depend on pulse history, not only instantaneous power, because residual heat increases the temperature sensitivity of the next conductance update.

H3. A coupling-aware schedule that separates high-current updates in space and time will reduce correlated weight error more effectively than uniform voltage compensation alone.

4. Methodology

4.1 Model Architecture

The study will use a calibrated reduced-order model to test thousands of pulse sequences without fabricated measurements. The main case is a 4-layer, 32 by 32 array. Smaller 8 by 8 by 4 structures support finite-element checks; 64 by 64 arrays with up to eight layers test scaling. The sweep covers 50, 100, and 200 nm cell pitches, published vertical dielectric thicknesses, and edge- and top-cooled boundaries [2], [8], [13].

Each word-line and bit-line segment will contain resistance and self-inductance. Mutual inductance will couple parallel segments within and between layers; capacitance will connect adjacent, crossing, and interlayer conductors. PEEC-style matrices generated from the geometry will be checked against published extraction procedures [4], [5]. The memristor will use a field- and temperature-dependent state variable x calibrated to RRAM switching curves [7]. The electrical solve provides local device voltage V and current I:

$$\frac{dx}{dt} = f(V, T, x), \quad G = G(x, T)$$

The thermal model will assign heat capacity to each active cell and thermal resistance to neighboring cells, electrodes, and the external sink. For cell i, the transient balance is:

$$C_i \frac{dT_i}{dt} = V_i I_i - \sum_j \frac{T_i - T_j}{R_{ij}} - \frac{T_i - T_{\text{amb}}}{R_{\text{i,sink}}}$$

A finite-element heat solve in FEniCSx will provide the reduced thermal matrix for small structures. Larger simulations will use it in ngspice or an equivalent Python modified-nodal-analysis solver. Electrical and thermal variables will be iterated within each time step until current and temperature converge.

4.2 Experimental Factors and Controls

Six factors will be varied: layer count (1, 2, 4, 8), lateral pitch (50-200 nm), pulse rise time (0.1-100 ns), duty cycle (1-50%), fraction of cells in the low-resistance state (10-90%), and effective sink resistance. Pulse amplitude and width will be normalized to produce the same intended single-cell conductance change, preventing energy from differing by design. Activity patterns will include isolated updates, adjacent simultaneous updates, checkerboard updates, and bursts in one layer.

Four ablation conditions will separate the mechanisms: ideal interconnect and fixed temperature; RLC parasitics with fixed temperature; electrothermal coupling with resistive lines only; and the full RLC-electrothermal model. A fifth will remove mutual inductance alone, isolating its relevance in the tested timing range.

4.3 Outcome Measures

State drift after intended updates stop will be measured with the Euclidean norm:

$$D(t) = \frac{\|G(t) - G(0)\|_2}{\|G(0)\|_2}$$

Other measures are nonselected-cell level changes, pulse overshoot, settling time, peak temperature, and spatial autocorrelation. Runaway begins when the dominant eigenvalue's real part crosses zero under a repeated pulse train, marking loss of a stable electrothermal fixed point. Peak temperature remains a separate measure.

4.4 Learning-Dynamics Test

The array model will be connected to a 784-128-10 multilayer perceptron trained on MNIST, a benchmark used in prior memristive hardware studies [10], [15]. Positive and negative weights will use differential conductance pairs. Each requested in-situ update will become programming pulses that pass through the coupled model before the optimizer receives the new conductance. The ideal software network, ideal-device crossbar, and non-ideal ablations will start from the same weights and data order.

Measures will include test accuracy, loss relative to the ideal-device crossbar, epochs needed to reach 95% of its final accuracy, energy per accepted update, and update error by layer and position. Twenty seeds will be used at each selected operating point. The aim is to measure how one training process changes across the drift and stability boundaries, not to maximize the MNIST score.

4.5 Sampling and Statistical Analysis

An initial Latin-hypercube sample of 250 configurations will cover the design space. A Gaussian-process surrogate will identify the stability boundary, where another 100 configurations will be added. Deterministic ablations will be followed by Monte Carlo runs with published ranges of resistance, threshold, and thermal-property variation. Variance-based sensitivity indices will estimate main effects and interactions; bootstrap 95% intervals will be reported for accuracy loss, drift, and boundary location.

For H1, the dependent variables will be victim-cell pulse error and unintended state change as functions of rise time, pitch, and mutual inductance. H2 will be tested by comparing equal-energy pulse trains with different spacing; evidence would be a different drift rate or fixed-point stability despite equal total energy. H3 will compare three controls at matched training work: uniform pulses, position-aware voltage correction, and a scheduler that avoids simultaneous updates in thermally and electromagnetically coupled neighborhoods. Using identical seeds in paired comparisons will reduce variation from the training sequence.

4.6 Calibration and Validation

Validation will precede the learning tests. The single-cell model will reproduce published switching and temperature sensitivity [7], [10]. The thermal solver will reproduce the reported layer ordering, transient heating, and pulse-memory effect [8], [9]. The RLC network will be checked against PEEC or S-parameter trends [4], [5], and the full simulator will repeat one published array-level case [3], [12]. Final sensitivity and learning runs will use held-out configurations.

4.7 Reproducibility, Safety, and Ethics

The project uses no human participants, animals, or personal data. Code, parameters, seeds, and generated data will be versioned. Figures will label reproduced results, calibration output, and new simulations. No numerical outcome is asserted in advance. Material-specific calibration will be identified, and broader claims will be limited to trends that survive parameter variation.

5. Expected Results and Interpretation

The expected output is a three-part operating map. One region should show reversible waveform error and little movement in unselected cells. A second should show recoverable, spatially correlated drift while the thermal fixed point remains stable. The last should show no stable periodic solution for the pulse train, marking a runaway-prone condition. These are hypotheses, not numerical findings.

Mutual inductance may prove negligible for most practical pulses. That would place a timing and geometry bound on its importance. If it matters only through temperature-dependent switching, the ablation should show a small waveform change but a larger state error. The result will indicate whether mitigation should target interconnect geometry, pulse shape, cooling, or update scheduling.

The design rule will be a safe operating envelope. For each layer count and cooling condition, it will give allowable combinations of edge rate, duty cycle, and simultaneous-update density for a chosen drift and accuracy budget. Mitigations will be compared by accuracy recovered per added energy and training time.

6. Innovation and Significance

The proposal joins electromagnetic coupling, device-state kinetics, 3D heat flow, and repeated learning updates. Its fixed-point test asks when the system stops returning to a stable cycle, rather than only how hot one case becomes. Recent coupled models cover large planar arrays and a small two-layer network [3], [12]; this study adds vertical heat paths, interlayer coupling, pulse history, and online updates.

7. Limitations

A reduced model cannot reproduce every filament shape, interface defect, or package heat path. Each calibrated device family will remain internally consistent. The small MNIST network exposes learning dynamics but does not represent a production model, and finite-element checks will be limited to small arrays. Scaling conclusions will carry uncertainty bands and remain model-based guidance.

8. Work Plan and Feasibility

Period	Main work	Deliverable
Months 1-2	Literature audit; device, RLC, and thermal model calibration	Validated unit models
Months 3-4	3D extraction, finite-element checks, and ablation simulations	Coupled simulator
Months 5-6	Adaptive parameter sweep and stability-boundary analysis	Operating maps
Month 7	MNIST in-situ learning tests and mitigation comparison	Learning results
Month 8	Uncertainty analysis, code release, and final report	Reproducible package

The open-source stack is Python, FEniCSx, ngspice, and PyTorch. If needed, the large sweep will use 32 by 32 by 8 while keeping all factors.

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